

Matrices- Operations and Applications

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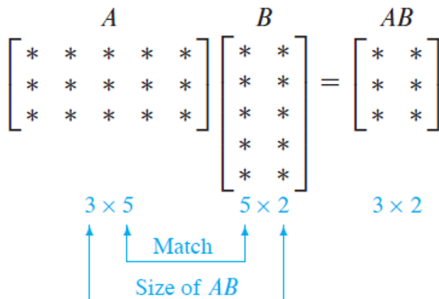
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Matrix Multiplication and Determinants

Matrix Multiplication: Definition and Rules

- Matrix multiplication combines two matrices A and B to produce C .
- Rules:
 - A must be $m \times n$ and B must be $n \times p$.
 - The resulting matrix C will be $m \times p$.
 - Element $C[i, j]$ is calculated as:

$$C[i, j] = \sum_{k=1}^n A[i, k] \cdot B[k, j].$$



Example 1: 2×2 Matrix Multiplication

Given:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

Compute $C = A \cdot B$:

$$C[1, 1] = 1 \cdot 5 + 2 \cdot 7 = 19,$$

$$C[1, 2] = 1 \cdot 6 + 2 \cdot 8 = 22.$$

$$C[2, 1] = 3 \cdot 5 + 4 \cdot 7 = 43,$$

$$C[2, 2] = 3 \cdot 6 + 4 \cdot 8 = 50.$$

Result:

$$C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

Example 2: 2×3 and 3×2 Matrix Multiplication

Given:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}.$$

Dimensions:

- A is 2×3 , B is 3×2 , C will be 2×2 .

Compute C :

$$\begin{aligned} C[1,1] &= 1 \cdot 7 + 2 \cdot 9 + 3 \cdot 11 = 58, & C[1,2] &= 1 \cdot 8 + 2 \cdot 10 + 3 \cdot 12 = 64. \\ C[2,1] &= 4 \cdot 7 + 5 \cdot 9 + 6 \cdot 11 = 139, & C[2,2] &= 4 \cdot 8 + 5 \cdot 10 + 6 \cdot 12 = 154. \end{aligned}$$

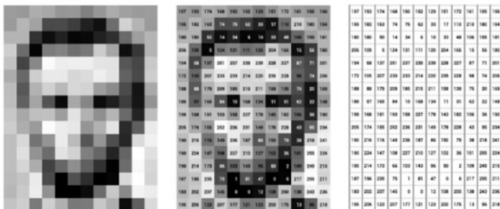
Result:

$$C = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}.$$

$$AB = \begin{matrix} \rightarrow \end{matrix} \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{matrix} \downarrow \\ \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix} \end{matrix} = \begin{bmatrix} \square & \square & 2(6) + 3(3) \\ \square & \square & \square \end{bmatrix} = \begin{bmatrix} \square & \square & 21 \\ \square & \square & \square \end{bmatrix}$$

$$\begin{matrix} \rightarrow \end{matrix} \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} \begin{matrix} \downarrow \\ \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix} \end{matrix} = \begin{bmatrix} \square & \square & 21 \\ \square & 1(3) + -5(-2) & \square \end{bmatrix} = \begin{bmatrix} \square & \square & 21 \\ \square & 13 & \square \end{bmatrix}$$

Image Processing: In image processing and computer vision, matrices are employed to represent images as grids of pixels and perform operations like blurring, sharpening, edge detection, and compression.



Digital image as matrix of numerical values. Numerical values represent pixel's brightness in this example.

Message: ATTACK IS TONIGHT

Hill Cipher

■ Encryption :

Cipher Text =(Plain Text x Key) Mod 26

■ Decryption:

Plain Text =(Cipher Text x Key⁻¹) Mod 26

■ Message: ATTACK IS TONIGHT

■ **Cipher Text = (Plain Text x Key) Mod 26**

$$= \begin{bmatrix} 0 & 19 & 19 \\ 0 & 2 & 10 \\ 8 & 18 & 19 \\ 14 & 13 & 8 \\ 6 & 7 & 19 \end{bmatrix} \times \begin{bmatrix} 3 & 10 & 20 \\ 20 & 9 & 17 \\ 9 & 4 & 17 \end{bmatrix} \text{ Mod } 26$$

$$\mathbf{Key} = \begin{bmatrix} 3 & 10 & 20 \\ 20 & 9 & 17 \\ 9 & 4 & 17 \end{bmatrix}$$

■ Message: ATTACK IS TONIGHT

Assign : A-Z 0-25

$$\begin{bmatrix} A & T & T \\ A & C & K \\ I & S & T \\ O & N & I \\ G & H & T \end{bmatrix} = \begin{bmatrix} 0 & 19 & 19 \\ 0 & 2 & 10 \\ 8 & 18 & 19 \\ 14 & 13 & 8 \\ 6 & 7 & 19 \end{bmatrix}$$

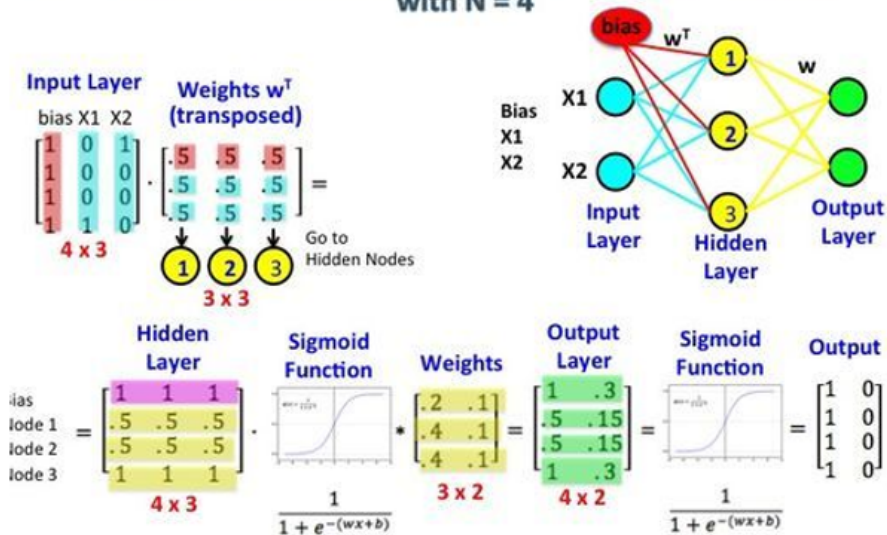
$$= \begin{bmatrix} 0 & 19 & 19 \\ 0 & 2 & 10 \\ 8 & 18 & 19 \\ 14 & 13 & 8 \\ 6 & 7 & 19 \end{bmatrix} \times \begin{bmatrix} 3 & 10 & 20 \\ 20 & 9 & 17 \\ 9 & 4 & 17 \end{bmatrix} \text{ Mod } 26$$

$$C_{11} = 0*3 + 19*20 + 19*9 = 551 \text{ Mod } 26 = 05$$

$$C_{12} = 0*10 + 19*9 + 19*4 = 247 \text{ Mod } 26 = 13$$

$$C_{13} = 0*20 + 19*17 + 19*17 = 646 \text{ Mod } 26 = 22$$

Color Guided Matrix Multiplication for a Binary Classification Task with N = 4



Determinants of 2×2 Matrices

Formula:

$$\det(A) = ad - bc \quad \text{for} \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Example:

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}.$$

Compute:

$$\det(A) = (3 \cdot 4) - (2 \cdot 1) = 12 - 2 = 10.$$

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

$$\det(A) = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)$$

$$\begin{aligned}
 |\mathbf{A}| &= aei + cdh + bgf - ceg - ahf - bdi \\
 &= a(ei - hf) - b(di - gf) + c(dh - eg) \\
 &= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\
 &= \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} - \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}
 \end{aligned}$$

Determinants of 3×3 Matrices

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

Compute:

$$\begin{aligned} \det(A) &= 1(5 \cdot 9 - 6 \cdot 8) - 2(4 \cdot 9 - 6 \cdot 7) + 3(4 \cdot 8 - 5 \cdot 7) \\ &= 1(-3) - 2(-6) + 3(-3) \\ &= -3 + 12 - 9 = 0. \end{aligned}$$

Result: $\det(A) = 0$ (Matrix is singular).

$$\begin{aligned} x_1 + 2x_2 &= 3 \\ 2x_1 + 5x_2 + 3x_3 &= 10 \\ x_2 + 2x_3 &= 5 \end{aligned} \quad A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 1 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 10 \\ 5 \end{bmatrix}$$

Matrix form: $A\mathbf{x} = \mathbf{b}$ Steps to find A^{-1} :

1. **Compute determinant:**

$$\det(A) = 1(5 \cdot 2 - 3 \cdot 1) - 2(2 \cdot 2 - 3 \cdot 0) + 0 = 7 - 8 = -1$$

2. **To get adjugate Transpose cofactor matrix C :**

$$C = \begin{bmatrix} 7 & -4 & 2 \\ -4 & 2 & -1 \\ 6 & -3 & 1 \end{bmatrix} \quad \text{adj}(A) = C^T = \begin{bmatrix} 7 & -4 & 6 \\ -4 & 2 & -1 \\ 2 & -1 & 1 \end{bmatrix}$$

3. **Compute inverse:**

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A) = \begin{bmatrix} -7 & 4 & -6 \\ 4 & -2 & 3 \\ -2 & 1 & -1 \end{bmatrix}$$

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -11 & 7 & -1 \end{bmatrix}^T$$

Cofactor Formula for Matrix Inverse

For an $n \times n$ matrix $A = [a_{ij}]$,

Minor M_{ij} :

- Delete row i and column j
- Determinant of the remaining $(n - 1) \times (n - 1)$ matrix

Cofactor C_{ij} :

$$C_{ij} = (-1)^{i+j} M_{ij}$$
$$\begin{bmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Cofactor matrix C and Adjugate matrix:

$$C = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix} \quad \text{adj}(A) = C^T \quad (\text{transpose of cofactor matrix})$$

Inverse formula:

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A) \quad \text{Note: } \det(A) \neq 0$$

Find the surface area of the cone

$$\mathbf{r}(r, \theta) = r \cos \theta \hat{i} + r \sin \theta \hat{j} + r \hat{k}, \quad 0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi.$$

Take $u = r$ and $v = \theta$. Surface Area of the cone would be

$$A = \iint_R |\mathbf{r}_r \times \mathbf{r}_\theta| dA = \int_0^{2\pi} \int_0^1 |\mathbf{r}_r \times \mathbf{r}_\theta| dr d\theta$$

Find the surface area of the cone

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$$\mathbf{r}_r = [\cos \theta, \sin \theta, 1], \quad \mathbf{r}_\theta = [-r \sin \theta, r \cos \theta, 0]$$

$$\mathbf{r}_r \times \mathbf{r}_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \theta & \sin \theta & 1 \\ -r \sin \theta & r \cos \theta & 0 \end{vmatrix} = ?$$

Find the surface area of the cone

$$\mathbf{r}(r, \theta) = r \cos \theta \hat{i} + r \sin \theta \hat{j} + r \hat{k}, \quad 0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi.$$

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Find the surface area of the cone

$$\mathbf{r}(r, \theta) = r \cos \theta \hat{i} + r \sin \theta \hat{j} + r \hat{k}, \quad 0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi.$$

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$$\mathbf{r}_r \times \mathbf{r}_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \theta & \sin \theta & 1 \\ -r \sin \theta & r \cos \theta & 0 \end{vmatrix} = ? \quad |\mathbf{r}_r \times \mathbf{r}_\theta| = ?$$

Then

$$A = \int_0^{2\pi} \int_0^1 |\mathbf{r}_r \times \mathbf{r}_\theta| dr d\theta = \sqrt{2}\pi$$

Find the surface area of a sphere of radius a .

$$\mathbf{r}(\theta, \phi) = (a \sin \phi \cos \theta) \hat{i} + (a \sin \phi \sin \theta) \hat{j} + a \cos \phi \hat{k},$$

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \pi$$

$$\mathbf{r}_\theta \times \mathbf{r}_\phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ f_\theta & g_\theta & h_\theta \\ f_\phi & g_\phi & h_\phi \end{vmatrix} = ? \quad |\mathbf{r}_\theta \times \mathbf{r}_\phi| = ?$$

Then

$$A = \int_0^\pi \int_0^{2\pi} |\mathbf{r}_\theta \times \mathbf{r}_\phi| \, d\theta \, d\phi = 4\pi a^2$$

Elementary Row Operations

Solve $-2x + 3y = -9$ and $x + y = 2$.

Elimination Method

$$-2x + 3y = -9 \quad (1)$$

$$x + y = 2 \quad (2)$$

Multiplying (2) by two and adding in (1), we get

$$y = -1,$$

and

$$x = 3$$

Thus $(3, -1)$ is the solution of equation.

Substitution Method

$$-2x + 3y = -9 \quad (3)$$

$$x + y = 2 \quad (4)$$

Substituting $x = 2 - y$ from (4) into (3), we get $2(2 - y) + 3y = -9$, which gives $y = -1$ and $x = 3$.

Thus $(3, -1)$ is the solution of equation.

Matrix Method

$$AX = B \text{ or } X = A^{-1}B$$

where

$$\underbrace{\begin{bmatrix} -2 & 3 \\ 1 & 1 \end{bmatrix}}_A, \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_X \& \underbrace{\begin{bmatrix} -9 \\ 2 \end{bmatrix}}_B$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -9 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

What Are Elementary Row Operations?

Simple operations on the rows of a matrix that transform it **without changing the solution set** of the corresponding linear system.

Applications

- Solving linear systems (Gaussian elimination)
- Finding matrix inverses
- Computing determinants
- Determining matrix rank

The Three Types of Elementary Row Operations

① **Row Switching** *Interchange two rows*

Notation: $R_i \leftrightarrow R_j$

② **Row Scaling** *Multiply a row by a **nonzero** scalar*

Notation: $R_i \leftarrow kR_i, \quad k \neq 0$

③ **Row Addition** *Add a multiple of one row to another row*

Notation: $R_i \leftarrow R_i + kR_j \quad (i \neq j)$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Row Switching

Swap R_1 and R_3 :

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\xrightarrow{R_1 \leftrightarrow R_3}$$

$$\begin{bmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

Notation: $R_1 \leftrightarrow R_3$

Row Scaling

Multiply R_2 by 2:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow 2R_2}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 8 & 10 & 12 \\ 7 & 8 & 9 \end{bmatrix}$$

Notation: $R_2 \leftarrow 2R_2$

Row Addition

Add $-4R_1$ to R_2 :

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 - 4R_1}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{bmatrix}$$

Notation:
 $R_2 \leftarrow R_2 - 4R_1$

Key Concepts: Row-Equivalent Matrices

Definition

Two matrices are **row-equivalent** if one can be obtained from the other through elementary row operations.

Row Echelon Forms

Using these operations, we can obtain:

- Row Echelon Form (REF)
- Reduced Row Echelon Form (RREF)

These forms reveal:

- Matrix rank
- Solutions to $A\mathbf{x} = \mathbf{b}$
- Linear dependence relations

Concept

Each row operation corresponds to **left-multiplication** by an **elementary matrix**.

Types:

- E_{ij} : switches rows i and j
- $E_i(k)$: multiplies row i by $k \neq 0$
- $E_{ij}(k)$: adds $k \times$ row j to row i

Example: For 3×3 :

$$E_{23}(5) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}$$

Multiplying $E_{23}(5)A$ adds $5 \times R_2$ to R_3 .

Row-Equivalent Matrices

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ -1 & 1 & 2 \end{bmatrix}$$

#	Operation	Result
1	$R_2 \leftarrow R_2 - 2R_1$	$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix}$
2	$R_3 \leftarrow R_3 + R_1$	$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{bmatrix}$
3	$R_3 \leftarrow R_3 - 3R_2$	$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Both matrices have the same:

- Row space
- Solution set for $Ax = 0$
- Rank (which is 3)

Conclusion: A and B are row-equivalent matrices.

Important Properties

Preserved

- Row space (basis may change)
- Solution set of $A\mathbf{x} = \mathbf{0}$ (null space)
- Linear independence of rows
- Matrix rank

May Change

- Column space
- Eigenvalues
- Determinant (affected by scaling and switching)

To solve $A\mathbf{x} = \mathbf{b}$

- 1 Form augmented matrix $[A|\mathbf{b}]$
- 2 Apply row operations to reach REF/RREF
- 3 Read solutions directly

To find A^{-1}

- 1 Form augmented matrix $[A|I]$
- 2 Apply row operations until $[I|A^{-1}]$
- 3 Extract inverse from right side

Find A^{-1} using Row Operations $[A|I]$

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 1 & 2 \end{bmatrix}$$

Start:

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

① $R_2 \leftarrow R_2 - 2R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

② $R_3 \leftarrow R_3 - R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & 0 & -1 & 2 & -1 & 1 \end{array} \right]$$

③ $R_3 \leftarrow -R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 & -1 \end{array} \right]$$

• $R_2 \leftarrow R_2 - 3R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 4 & -2 & 3 \\ 0 & 0 & 1 & -2 & 1 & -1 \end{array} \right]$$

• $R_1 \leftarrow R_1 - 2R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -7 & 4 & -6 \\ 0 & 1 & 0 & 4 & -2 & 3 \\ 0 & 0 & 1 & -2 & 1 & -1 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} -7 & 4 & -6 \\ 4 & -2 & 3 \\ -2 & 1 & -1 \end{bmatrix}$$

Elementary row operations are **fundamental tools** in linear algebra that allow us to:

- Simplify matrices systematically
- Solve linear equations efficiently
- Understand matrix properties
- Connect algebraic and geometric interpretations

They form the **computational backbone** of many linear algebra algorithms.

Suppose that A is an $n \times n$ matrix with entries in $\mathbb{R}$. Then the following statements are equivalent:

- a) A can be reduced to I_n by elementary row operations.
- b) A is invertible.
- c) $\det A \neq 0$.
- d) The system $Ax = 0$ has only the trivial solution.
- e) The system $Ax = b$ is soluble for every $b \in \mathbb{R}^n$.
- f) A has rank n .
- g) The rows of A are linearly independent.
- h) The columns of A are linearly independent.